Highlights

Stablecoins and the Bond Market: Yield-Curve Effects, Volatility Spillovers, Credit Spreads and Systemic Risk

稳定币与债券市场：收益率曲线效应、波动溢出、信用利差与系统性风险

Author Name

- A \$1 bn five-day stablecoin net issuance lowers 1M/3M Treasury yields by up to 3.4/2.4 bps within three weeks.
- 10 亿美元的 5 日稳定币净发行在三周内使 1M/3M 国债收益率最多下行 3.4/2.4 个基点。
- Yield effects concentrate at the short end; maturities beyond 2 years show no significant response.
- 收益率效应集中于短端，2 年期以上期限无显著响应。
- Stablecoin–bond connectedness is 19.7% on average; stablecoins turn into net transmitters in crisis windows.
- 稳定币与债市的平均连通性为 19.7%；危机窗口内稳定币转为波动净输出者。
- USDC flows dominate the yield effect; redemption effects exceed issuance effects in magnitude.
- USDC 流量主导收益率效应；赎回效应在量级上超过发行效应。
- A 10% USDT run raises the 3-month yield by 47–60 bps in calibrated stress tests.
- 校准压力测试显示，10% 的 USDT 挤兑将推高 3 个月期收益率 47–60 个基点。

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Abstract

Stablecoin issuers have become structural marginal buyers of U.S. money-market and short-dated fixed-income assets. This paper studies how stablecoin issuance and redemption affect the bond market along four dimensions: the Treasury yield curve, volatility spillovers, credit spreads (as a proxy for the commercial paper market, for which direct data are unavailable), and systemic risk. Using daily data from 2020-01 to 2026-06 (1,617 trading days) covering the combined market capitalization of USDT, USDC, DAI and BUSD, FRED Treasury yields and money-market variables, on-chain transfer and exchange flows, and crypto derivatives indicators, we deploy local projections (LP), LP with Elastic-Net residual instruments (LP-IV), vector autoregressions, Diebold–Yilmaz and rolling time-varying connectedness measures, two-stage least squares with credit-spread proxies, event studies, Rigobon (2003) heteroskedasticity identification, and calibrated stress tests. Local projections show that a \$1 bn five-day net issuance lowers the 1-month Treasury yield by 0.23 bps on impact and by up to 3.4 bps cumulatively after about three weeks, with the 3-month yield falling by up to 2.4 bps; effects are statistically and economically insignificant beyond the 2-year maturity, consistent with reserve portfolios concentrated at the short end. The instrumental-variable stage delivers a strong first stage for the DeFi-TVL residual instrument ($F = 13.2$) but imprecise second-stage estimates, so we treat LP as the preferred specification and IV estimates as

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bounds. Full-sample connectedness between the stablecoin market and bond-market variables is 19.7%; the stablecoin market is approximately neutral on average (net spillover −0.1 pp) but turns into a net transmitter in crisis windows (+1.7 pp vs. −2.4 pp in normal times). Event studies around the USDC depeg show money-market stress rising (+3 bps on the SOFR–RRP spread, $p = 0.045$) and VIX up 7.4 points within five days, while the 3-month yield falls 32 bps over ten days, indicating flight-to-quality. Heterogeneity results reveal that USDC flows drive the yield effect (−2.19 bps per $1 bn) and that redemption effects (−3.2 bps per $1 bn outflow) exceed issuance effects (−0.5 bps) in magnitude. A calibrated stress test implies that a 10% USDT run ($18.6 bn forced sales) would raise the 3-month yield by 47–60 bps and high-yield spreads by about 57 bps; a $50 bn joint run would raise the 3-month yield by 126–161 bps. The results support reserve-composition regulation in the spirit of the GENIUS Act while warning that forcing reserves into bills tightens the stablecoin–Treasury coupling.

稳定币发行方已成为美国货币市场与短期固定收益资产的结构性边际买方。本文从四个维度研究稳定币发行与赎回对债券市场的影响：国债收益率曲线、波动溢出、信用利差（作为数据不可得的商业票据市场的代理）以及系统性风险。基于 2020 年 1 月至 2026 年 6 月（1,617 个交易日）的日度数据——涵盖 USDT、USDC、DAI、BUSD 合计市值、FRED 国债收益率与货币市场变量、链上转账与交易所流量以及加密衍生品指标——本文综合运用局部投影（LP）、Elastic Net 残差工具变量局部投影（LP-IV）、向量自回归、Diebold–Yilmaz 与滚动时变连通性测度、基于信用利差代理的两阶段最小二乘、事件研究、Rigobon（2003）异方差识别与校准压力测试。局部投影显示，10 亿美元的 5 日净发行在当期使 1 个月期国债收益率下行 0.23 个基点，约三周后累计最多下行 3.4 个基点，3 个月期收益率最多下行 2.4 个基点；2 年期以上期限的影响在统计与经济意义上均不显著，与储备组合集中于短端的现实一致。工具变量估计中，DeFi-TVL 残差工具的第一阶段较强（$F = 13.2$），但第二阶段估计不精确，因此本文以 LP 为优选规格、以 IV 估计为区间参考。稳定币市场与债市变量的全样本总连通性为 19.7%；稳定币市场平均而言近似中性（净溢出 −0.1 个百分点），但在危机窗口转为净输出者（危机期 +1.7 个百分点，正常期 −2.4 个百分点）。USDC 脱锚事件研究显示货币市场压力上升（SOFR–RRP 利差走阔 3 个基点，$p = 0.045$）、VIX 五日内上升 7.4 个点，而 3 个月期收益率十日内下行 32 个基点，呈避险特征。异质性结果表明，USDC 流量主导了收益率效应（每 10 亿美元 −2.19 个基点），且赎回效应（每 10 亿美元流出 −3.2 个基点）在量级上超过发行效应（−0.5 个基点）。校准压力测试显示，10% 的 USDT 挤兑（186 亿美元被迫抛售）将推高 3 个月期收益率 47–60 个基

点、高收益利差约 57 个基点；500 亿美元联合挤兑将推高 3 个月期收益率 126–161 个基点。研究结果支持 GENIUS Act 式的储备结构监管，同时警示强制储备集中于国库券会收紧稳定币与国债市场之间的耦合。

Keywords: Stablecoins, Treasury market, Local projections, Instrumental variables, Spillovers, Credit spreads, Systemic risk, 稳定币, 国债市场, 局部投影, 工具变量, 溢出效应, 信用利差, 系统性风险, *JEL classification*: E44, G12, G15, G23

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1. Introduction / 引言

Stablecoins have grown from a niche settlement token into a structurally important class of money-market investors. The combined market capitalization of the four largest U.S.-dollar stablecoins (USDT, USDC, DAI and BUSD) rose from about \$4.8 bn in January 2020 to \$262.8 bn by the end of June 2026, a more than fifty-fold expansion, with Tether alone holding roughly \$184.7 bn of reserve assets. Because issuers back their tokens predominantly with short-dated Treasuries, overnight repos, money-market funds and—historically—commercial paper, every dollar of net stablecoin issuance or redemption translates into a corresponding purchase or sale at the front end of the U.S. rates complex. Stablecoins are therefore no longer merely a crypto-ecosystem plumbing issue: they are a marginal price setter in the very markets where the U.S. government funds itself and where corporations obtain short-term credit.

稳定币已从一种小众的结算代币成长为结构性重要的货币市场投资者类别。四大美元稳定币（USDT、USDC、DAI 与 BUSD）的合计市值从 2020 年 1 月的约 48 亿美元增长到 2026 年 6 月末的 2,628 亿美元，扩张超过五十倍，仅 Tether 一家就持有约 1,847 亿美元的储备资产。由于发行方主要以短期国债、隔夜回购、货币市场基金以及（历史上的）商业票据为其代币提供储备支持，每一美元的稳定币净发行或净赎回都会转化为美国利率体系短端相应的买入或卖出。因此，稳定币不再仅仅是加密生态的内部管道问题：它已成为美国政府融资市场与企业短期信用市场上的边际定价者。

Despite this institutional reality, the empirical bond-market footprint of stablecoins remains incompletely mapped. **(author?)** [1] provide the first causal estimates of stablecoin flows on safe-asset prices, and **(author?)** [2] identify stablecoin shocks with narrative and heteroskedasticity-based strategies. This paper deliberately does not replicate either study; instead, it borrows their local-projection and identification toolkit and extends the research design to three connected markets—the Treasury market, the commercial paper market (proxied by credit and money-market spreads because direct commercial-paper data are unavailable), and cross-market systemic-risk contagion—while emphasizing an asymmetry that the literature has not yet quantified jointly: issuance-driven allocation effects and redemption-driven fire-sale effects work in opposite directions and need not be of equal magnitude.

尽管存在上述制度现实，稳定币在债券市场上的实证足迹仍不完整。**(author?)** [1] 首次给出稳定币流量对安全资产价格的因果估计，**(author?)** [2] 则利用叙事识别与异方差识别策略识别稳定币冲击。本文刻意不复刻这两项研究，而是借鉴其局部投影与识别工具箱，将研究设计拓展到三个相互关联的市场——国债市场、商业票据市场（因直接的商业票据数据不可得，用信用利差与货币市场利差代理）以及跨市场系统性风险传染——并强调文献尚未联合量化的一种非对称性：发行驱动的配置效应与赎回驱动的抛售效应方向相反，且量级未必相等。

We ask four questions. *First* (Q1), how do stablecoin issuance and redemption affect the Treasury yield curve, and is the effect concentrated at the short end where reserves are actually invested? *Second* (Q2), how strong is the volatility connectedness between the stablecoin market and the bond market, and does the direction of net spillovers change in crises? *Third* (Q3), how does stablecoin reserve allocation transmit to credit spreads, which proxy for the commercial paper market? *Fourth* (Q4), would a stablecoin run constitute a systemic event for the bond market, and how large would the impact be under calibrated stress scenarios?

本文提出四个问题。第一（Q1），稳定币发行与赎回如何影响国债收益率曲线？该效应是否集中于储备实际配置的短端？第二（Q2），稳定币市场与债市之间的波动连通性有多强？净溢出方向在危机时期是否发生转变？第三（Q3），稳定币储备配置如何传导至信用利差（商业票据市场的代理变量）？第四（Q4），稳定币挤兑是否会对债市构成系统性事件？在校准压力情景下影响有多大？

Our answers rest on a deliberately multi-method design estimated on daily data from January 2020 to June 2026. For Q1 we estimate local projections [3] of cumulative yield changes on five-day stablecoin net flows across seven maturities, complement them with local-projection instrumental-variable estimates using Elastic-Net residual instruments in the spirit of **(author?)** [1], and cross-check with a Cholesky-ordered VAR. For Q2 we compute the **(author?)** [4] spillover index and its rolling time-varying counterpart in the tradition of **(author?)** [5]. For Q3 we estimate two-stage least squares regressions of investment-grade and high-yield OAS, the TED spread and the SOFR–ON-RRP spread on stablecoin flows, and run event studies [6] around five stablecoin-specific events. For Q4 we apply **(author?)** [7] heteroskedasticity identification to crisis vs. normal covariance regimes and calibrate four stress scenarios—a 10% USDT run, an SVB-like USDC depeg, a $50 bn joint run and a regulatory reserve rebalancing—to the estimated impulse responses.

我们的答案建立在日度数据（2020 年 1 月至 2026 年 6 月）上的多方法设计。针对 Q1，本文估计七个期限上累计收益率变动对稳定币 5 日净流量的局部投影 [3]，辅以借鉴 **(author?)** [1] 的 Elastic Net 残差工具变量局部投影估计，并用 Cholesky 排序的 VAR 交叉验证。针对 Q2，本文计算 **(author?)** [4] 溢出指数及其沿袭 **(author?)** [5] 传统的滚动时变对应指标。针对 Q3，本文对投资级与高收益 OAS、TED 利差和 SOFR–ON-RRP 利差关于稳定币流量做两阶段最小二乘估计，并围绕五个稳定币特有事件开展事件研究 [6]。针对 Q4，本文将 **(author?)** [7] 异方差识别应用于危机期与正常期的协方差区制，并把四个压力情景——10% 的 USDT 挤兑、类 SVB 的 USDC 脱锚、500 亿美元联合挤兑与监管强制储备再平衡——校准到估计的脉冲响应上。

Four findings emerge. *First*, stablecoin flows move the front end of the curve in the expected direction: a $1 bn five-day net issuance lowers the 1-month yield by 0.23 bps on impact and by up to 3.4 bps after about three weeks, and lowers the 3-month yield by up to 2.4 bps, with no significant effect at or beyond the 2-year maturity. *Second*, average connectedness between stablecoins and bond-market variables is moderate (19.7%), but its direction is state-dependent: the stablecoin market is a net receiver of spillovers on average yet turns into a net transmitter in crisis windows. *Third*, the credit-spread channel is detectable in event time—the USDC depeg raised money-market stress and VIX while lowering bill yields through flight-to-quality—but regression-based spread effects are imprecise, which we interpret as evidence that daily spread variation is dominated by aggregate risk factors rather than stablecoin flows alone. *Fourth*, calibrated stress tests imply economically meaningful systemic footprints: a 10% USDT redemption would raise the 3-month yield by 47–60 bps, and a $50 bn joint run by 126–161 bps, under linear-extrapolation caveats.

本文得到四点主要发现。第一，稳定币流量以预期方向影响收益率曲线短端：10 亿美元的 5 日净发行在当期使 1 个月期收益率下行 0.23 个基点，约三周后最多下行 3.4 个基点，并使 3 个月期收益率最多下行 2.4 个基点，而 2 年期及以上期限无显著效应。第二，稳定币与债市变量之间的平均连

通性中等（19.7%），但其方向具有状态依赖性：稳定币市场平均是溢出的净接收者，但在危机窗口转为净输出者。第三，信用利差渠道在事件时间维度上可识别——USDC 脱锚推高了货币市场压力与 VIX，同时通过避险需求压低国库券收益率——但基于回归的利差效应不够精确，本文将其解读为日度利差变动由总体风险因子主导、而非仅由稳定币流量驱动的证据。第四，校准压力测试显示了具有经济意义的系统性足迹：在线性外推的注意事项下，10% 的 USDT 赎回将推高 3 个月期收益率 47–60 个基点，500 亿美元联合挤兑将推高 126–161 个基点。

The paper contributes to three strands of literature. Relative to the stablecoin–safe-asset literature [1, 2], we add the credit-spread dimension, the issuance-redemption asymmetry, and a calibrated systemic-risk quantification in a single coherent framework. Relative to the stablecoin run literature [8, 9], we provide daily-frequency evidence on how runs transmit into money-market stress and bill yields. Relative to the connectedness literature [4, 5], we document the sign switches of stablecoin net spillovers around crypto-specific and banking-specific crises. The findings speak directly to the reserve-composition mandates of the GENIUS Act [10]: forcing reserves into short Treasuries improves individual-coin safety but deepens the structural coupling between stablecoins and the bill market, with quantifiable consequences for yield volatility under run scenarios.

本文对三支文献有所贡献。相对于稳定币—安全资产文献 [1, 2]，本文在统一框架内加入了信用利差维度、发行—赎回非对称性与校准的系统性风险量化。相对于稳定币挤兑文献 [8, 9]，本文提供了挤兑如何传导至货币市场压力与国库券收益率的日度频率证据。相对于连通性文献 [4, 5]，本文记录了稳定币净溢出在加密特有与银行业危机前后的符号转换。研究结果直接回应 GENIUS Act [10] 的储备结构强制要求：强制储备集中于短期国债虽提升个体稳定币的安全性，却加深了稳定币与国库券市场的结构性耦合，并对挤兑情景下的收益率波动产生可量化的影响。

The remainder of the paper is organized as follows. Section 2 provides institutional background and develops testable hypotheses. Section 3 describes the data. Section 4 lays out the methodology. Section 5 reports the empirical results for Q1–Q4. Section 6 presents heterogeneity and robustness analyses. Section 7 discusses economic and policy implications. Section 8 concludes.

本文其余部分安排如下。第 2 节介绍制度背景并提出可检验假说。第 3 节描述数据。第 4 节阐述方法。第 5 节报告 Q1–Q4 的实证结果。第 6 节给出异质性与稳健性分析。第 7 节讨论经济与政策含义。第 8 节总结全文。

2. Institutional Background and Hypotheses / 制度背景与研究假说

2.1. Stablecoin reserve mechanics / 稳定币储备机制

Fiat-backed stablecoins operate a narrow-bank-like balance sheet: tokens are issued against dollar inflows and the proceeds are invested in reserve assets. Tether's and Circle's disclosed portfolios are dominated by short-dated Treasury bills, overnight reverse repo and money-market instruments, while Tether historically held substantial non-financial commercial paper before committing to eliminate it in July 2022. Two accounting identities follow. First, net issuance mechanically increases demand for the assets in the reserve basket; net redemption mechanically forces their sale. Second, because the basket is concentrated at maturities of three months or less, the demand effect should load on the shortest tenors of the yield curve and decay with maturity—a preferred-habitat logic in the sense of **(author?)** [11], with stablecoin issuers as a habitat investor in bills. Treasury demand is also interest-elastic in aggregate [12], so a new large buyer compresses short-end yields unless fully offset by supply.

法币抵押型稳定币的资产负债表类似狭义银行：代币针对美元流入发行，所得资金投资于储备资产。Tether 与 Circle 披露的组合以短期国库券、隔夜逆回购与货币市场工具为主，而 Tether 在 2022 年 7 月承诺清零之前曾持有大量非金融商业票据。由此得到两个会计恒等式。第一，净发行在机械意义上增加储备篮子内资产的需求；净赎回则强制其出售。第二，由于储备篮子集中于三个月及以内期限，需求效应应主要作用于收益率曲线的最短端并随期限衰减——这是 **(author?)** [11] 意义上的偏好栖息地逻辑，稳定币发行方充当国库券市场上的栖息地投资者。国债需求在总量上也是利率弹性的 [12]，因此除非被供给完全对冲，新的大型买方会压缩短端收益率。

2.2. Two-directional transmission / 双向传导机制

The transmission is deliberately two-directional. In the expansion scenario, net issuance raises reserve-allocation demand, which lowers short-end yields and compresses credit spreads. In the run scenario, redemptions force fire sales of reserve assets, which push short-end yields up, widen credit spreads and impair bond-market liquidity, potentially feeding back into money market funds and banks that share the same asset pool [8, 9]. Because sales under duress are executed into falling markets with limited dealer intermediation, the redemption leg should be stronger per dollar than the issuance leg—an asymmetry we test directly.

传导机制在设计上就是双向的。在扩张情景下，净发行提高储备配置需求，从而压低短端收益率并压缩信用利差。在挤兑情景下，赎回迫使储备资产被抛售，推高短端收益率、走阔信用利差并损害债市流动性，并可能反馈到共享同一资产池的货币市场基金与银行 [8, 9]。由于压力下的抛售是在下跌市场中、做市商中介能力有限的条件下执行的，赎回腿每美元的效应应强于发行腿——本文将直接检验这一非对称性。

2.3. Testable hypotheses / 可检验假说

For the Treasury market (Q1) we test: H1a, net issuance lowers short-end yields ($\beta_h < 0$ at the 1M and 3M tenors); H1b, redemption or run shocks raise short-end yields ($\beta_h > 0$ in crisis subsamples or for outflow shocks); H1c, redemption effects exceed issuance effects in magnitude ($|\beta_{\text{outflow}}| > |\beta_{\text{inflow}}|$); H1d, effects concentrate at the short end and transmit to the long end only indirectly through term premia (significant at 1M/3M, weak or delayed at 10Y).

针对国债市场（Q1），本文检验：H1a，净发行压低短端收益率（1M 与 3M 期限上 $\beta_h < 0$）；H1b，赎回或挤兑冲击推高短端收益率（危机子样本或流出冲击下 $\beta_h > 0$）；H1c，赎回效应在量级上超过发行效应（$|\beta_{\text{outflow}}| > |\beta_{\text{inflow}}|$）；H1d，效应集中于短端，仅通过期限溢价间接传导至长端（1M/3M 显著，10Y 较弱或滞后）。

For volatility spillovers (Q2): H2a, the total spillover index between stablecoin and bond markets is significantly positive; H2b, connectedness rises in crises such as the Terra collapse and the SVB episode; H2c, the stablecoin market switches from a net receiver to a net transmitter of volatility in crisis windows.

针对波动溢出（Q2）：H2a，稳定币与债市之间的总溢出指数显著为正；H2b，在 Terra 崩盘与 SVB 事件等危机期间连通性上升；H2c，稳定币市场在危机窗口内从波动的净接收者转为净输出者。

For credit spreads (Q3): H3a, stablecoin accumulation of credit assets compresses credit spreads ($\beta < 0$); H3b, regulatory events that force reserve rebalancing away from risky assets widen credit spreads in event windows (CAR > 0); H3c, investment-grade and high-yield spreads respond differently.

针对信用利差（Q3）：H3a，稳定币增持信用资产压缩信用利差（$\beta < 0$）；H3b，迫使储备减持风险资产的监管事件在事件窗口内使信用利差走阔（CAR > 0）；H3c，投资级与高收益利差的响应不同。

For systemic risk (Q4): H4a, stablecoin crisis shocks significantly raise short-end yields and stress spreads under heteroskedasticity identification;

H4b, under stress scenarios, cross-market contagion is economically significant, with simultaneous responses of bills, credit and volatility.

针对系统性风险（Q4）：H4a，在异方差识别下，稳定币危机冲击显著推高短端收益率与压力利差；H4b，在压力情景下，跨市场传染具有经济显著性，国库券、信用与波动率同时响应。

3. Data and Variables / 数据与变量

3.1. Data sources / 数据来源

The sample runs from 2020-01-01 to 2026-06-30. All series are drawn from four local datasets. First, FRED daily bond-market data [13] provide seven Treasury tenors (DGS1MO, DTB3, DGS1, DGS2, DGS5, DGS10, DGS30), curve spreads (T10Y2Y, T10Y3M), monetary and liquidity variables (DFF, SOFR, RRPONTSYD, WALCL), credit spreads (BAMLC0A0CM, BAMLH0A0HYM2, TEDRATE) and controls (VIXCLS, DTWEXBGS, DCOILWTICO). Second, a merged macro–crypto panel supplies the S&P 500, gold, DeFi TVL, Bitcoin, the dollar index, the SOFR–ON-RRP spread, policy uncertainty and the USDT OTC premium. Third, daily supply series for USDT, USDC, DAI and BUSD are combined into total stablecoin market capitalization. Fourth, on-chain indicators (transfer volume, exchange inflows and outflows, active addresses), crypto-wide market metrics (total market capitalization, open interest) and Bitcoin derivatives (funding rates, liquidations, leverage, the Fear & Greed Index) complete the information set.

样本期为 2020 年 1 月 1 日至 2026 年 6 月 30 日。所有序列来自四个本地数据集。第一，FRED 日度债券市场数据 [13] 提供七个国债期限（DGS1MO、DTB3、DGS1、DGS2、DGS5、DGS10、DGS30）、曲线利差（T10Y2Y、T10Y3M）、货币与流动性变量（DFF、SOFR、RRPONTSYD、WALCL）、信用利差（BAMLC0A0CM、BAMLH0A0HYM2、TEDRATE）以及控制变量（VIXCLS、DTWEXBGS、DCOILWTICO）。第二，宏观—加密合并面板提供标普 500、黄金、DeFi TVL、比特币、美元指数、SOFR–ON-RRP 利差、政策不确定性与 USDT 场外溢价。第三，USDT、USDC、DAI 与 BUSD 的日度供应序列合成稳定币总市值。第四，链上指标（转账量、交易所流入流出、活跃地址）、加密市场全局指标（总市值、未平仓合约）与比特币衍生品指标（资金费率、爆仓量、杠杆率、恐惧贪婪指数）构成完整信息集。

Three data limitations bind and are addressed transparently. First, no six-month Treasury series is available; the tenor grid is 1M, 3M, 1Y, 2Y, 5Y, 10Y and 30Y. Second, commercial paper rates (CP-OIS) are unavailable in the local data; the ICE BofA OAS series start only in June 2023 and the TED spread was discontinued in January 2022. We therefore proxy the CP/credit channel with investment-grade and high-yield OAS (post-2023 subsample), the TED spread (2020–2022 subsample) and the SOFR–ON-RRP spread (full sample), and state this proxy strategy wherever it is used. Third, the supply series contains occasional revision jumps (e.g. a spurious +34% print in April 2020); growth rates are winsorized at the 0.5/99.5 percentiles.

三项数据约束存在并被透明处理。第一，没有 6 个月期国债序列，期限网格为 1M、3M、1Y、2Y、5Y、10Y 与 30Y。第二，本地数据中没有商业票据利率（CP-OIS）；ICE BofA 的 OAS 序列仅从 2023 年 6 月开始，TED 利差已于 2022 年 1 月停更。因此本文用投资级与高收益 OAS（2023 年后子样本）、TED 利差（2020–2022 年子样本）和 SOFR–ON-RRP 利差（全样本）代理 CP/信用渠道，并在使用处明确说明这一代理策略。第三，供应序列存在个别修订跳点（如 2020 年 4 月的 +34% 虚假观测）；增长率在 0.5/99.5 分位缩尾。

3.2. Variable construction / 变量构造

Total stablecoin market capitalization is $MC_t = USDT_t + USDC_t + DAI_t + BUSD_t$. The main regressors are the one-day net issuance $\Delta Issuance_t = MC_t - MC_{t-1}$ and the five-day net flow $Flow_t^{(5d)} = MC_t - MC_{t-5}$, both in billions of U.S. dollars; issuer-specific five-day flows for USDT and USDC are constructed analogously. Dependent variables are cumulative yield changes $\Delta y^{\tau}_{t+h} = y^{\tau}_{t+h} - y^{\tau}_{t-1}$ in basis points. Controls include changes in the federal funds rate and SOFR, VIX changes, dollar log returns and S&P 500 log returns. On-chain controls are the combined USDT+USDC transfer-volume log change, active-address log change and exchange net-flow change; derivatives controls are the funding rate, log liquidations, the estimated leverage ratio and the Fear & Greed Index.

稳定币总市值定义为 $MC_t = USDT_t + USDC_t + DAI_t + BUSD_t$。核心解释变量为日度净发行 $\Delta Issuance_t = MC_t - MC_{t-1}$ 与 5 日净流量 $Flow_t^{(5d)} = MC_t - MC_{t-5}$，单位均为十亿美元；USDT 与 USDC 的发行人层面 5 日流量按同法构造。被解释变量为累计收益率变动 $\Delta y^{\tau}_{t+h} = y^{\tau}_{t+h} - y^{\tau}_{t-1}$，单位为基点。控制变量包括联邦基金利率与 SOFR 的变动、VIX 变动、美元对数收益与标普 500 对数收益。链上控制变量为 USDT+USDC 合计转账量对数变动、活跃地址对数变动与交易所净流入变动；衍生品控制变量为资金费率、爆仓量对数、估计杠杆率与恐惧贪婪指数。

3.3. Summary statistics and stylized facts / 描述性统计与典型事实

Table 1 reports summary statistics. Stablecoin market capitalization averages \$136 bn over the sample with wide dispersion; the five-day net flow averages +\$0.55 bn with a standard deviation of \$1.26 bn and extremes of −\$4.9 bn and +\$7.4 bn, confirming that both large issuances and large redemptions occur in the sample. Figure 1 plots the joint evolution of stablecoin market capitalization, 3M/10Y yields, corporate OAS and the SOFR–RRP spread, with the four key events marked: the Terra/UST collapse (2022-05-09), Tether's zero-commercial-paper commitment (2022-07-27), the USDC depeg during the SVB failure (2023-03-10) and the Senate passage of the GENIUS Act (2025-06-17).

表 1 报告描述性统计。稳定币市值样本均值为 1,362 亿美元，离散度较大；5 日净流量均值为 +5.5 亿美元，标准差 12.6 亿美元，极值为 −49 亿美元与 +74 亿美元，表明样本内既存在大规模发行也存在大规模赎回。图 1 绘制稳定币市值、3M/10Y 收益率、公司债 OAS 与 SOFR–RRP 利差的联合演化，并标注四个关键事件：Terra/UST 崩盘（2022-05-09）、Tether 零商业票据承诺（2022-07-27）、SVB 倒闭期间的 USDC 脱锚（2023-03-10）以及 GENIUS Act 参议院通过（2025-06-17）。

Table 1: 描述性统计 / Summary statistics

Variable 变量	Mean 均值	Std 标准差	Min 最小值	Max 最大值	N
MC (stablecoin market cap, \$bn)	136.25	75.19	4.80	272.51	2363
$Flow^{(5d)}$ (\$bn)	0.55	1.26	−4.93	7.44	2358
ΔMC winsorized (%/day)	0.15	0.42	−0.82	2.35	2362
y^{1M} (%)	2.85	2.22	0.00	6.02	1617
y^{3M} (%)	2.81	2.11	−0.05	5.36	1617
y^{10Y} (%)	3.05	1.40	0.52	4.98	1617
Δy^{3M} (bp/day)	−0.44	2.60	−22.00	17.00	1262
ΔVIX (pts/day)	−0.09	2.08	−18.71	21.57	1311
ΔSOFR–RRP (bp/day)	0.00	2.07	−44.00	28.00	2362
ΔIG OAS (bp/day, 2023-06+)	−0.09	1.39	−4.00	10.00	622
ΔHY OAS (bp/day, 2023-06+)	−0.11	7.30	−32.00	59.00	623

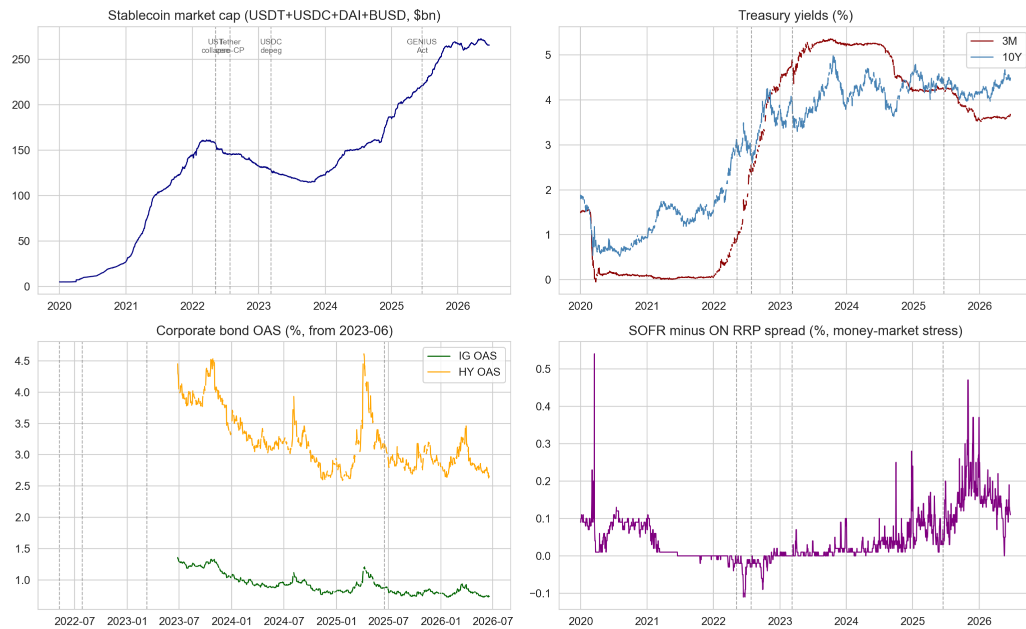

Figure 1: 典型事实：稳定币市值、国债收益率、信用利差与货币市场压力 / Stylized facts: stablecoin market cap, Treasury yields, credit spreads and money-market stress. Vertical dashed lines mark the UST collapse, Tether's zero-CP commitment, the USDC depeg and the GENIUS Act Senate passage. 竖虚线标注 UST 崩盘、Tether 零 CP 承诺、USDC 脱锚与 GENIUS Act 参议院通过。

4. Methodology / 方法

4.1. *Local projections and LP-IV for the yield curve* / 收益率曲线的局部投影与 *LP-IV*

The baseline specification for Q1 is the local projection [3]

$$\Delta y^{\tau}_{t+h} = \alpha^{\tau}_h + \beta^{\tau}_h \cdot Flow^{(5d)}_t + \sum_k \gamma^{\tau}_{h,k} X_{k,t} + \varepsilon^{\tau}_{t+h}, \qquad h = 0, 1, \ldots, 20, \quad (1)$$

estimated separately for each tenor $\tau \in \{1M, 3M, 1Y, 2Y, 5Y, 10Y, 30Y\}$ with Newey–West standard errors (lag $h + 2$). The sequence $\{\beta^{\tau}_h\}$ traces the impulse response of the yield at tenor τ to a one-billion-dollar five-day net issuance. Controls X comprise ΔDFF, $\Delta SOFR$, ΔVIX, $\Delta \ln DXY$ and $\Delta \ln SP500$.

Q1 的基准模型是局部投影 [3]，对每个期限 $\tau \in \{1M$, $3M$, $1Y$, $2Y$, $5Y$, $10Y$, $30Y\}$ 分别估计式 (1)，并使用 Newey–West 标准误（滞后 $h+2$）。系数序列 $\{\beta^{\tau}_h\}$ 刻画期限 τ 的收益率对十亿美元 5 日净发行的脉冲响应。控制变量 X 包括 ΔDFF、$\Delta SOFR$、ΔVIX、$\Delta \ln DXY$ 与 $\Delta \ln SP500$。

Because yields feed back into issuance incentives—higher bill yields raise reserve income and attract supply—the flow variable is endogenous. We follow **(author?)** [1] and construct Elastic-Net residual instruments: we regress crypto-native variables on the traditional-finance information set with a cross-validated elastic net and keep the residuals, which isolate crypto-specific variation orthogonal to bond-market conditions. Concretely, IV1 is the residual of Bitcoin log returns on the controls $\Delta \ln SP500$, $\Delta \ln DXY$, $\Delta \ln Gold$, ΔVIX, ΔDFF and Δy^{10Y}; IV2 is the analogous DeFi-TVL residual; IV3 is the demeaned USDT OTC premium. The first stage is $Flow^{(5d)}_t = \pi_0 + \pi_1 Z_t + \Pi X_t + u_t$ and the second stage replaces $Flow$ by $\widehat{Flow}$ in equation (1), estimated by 2SLS with Bartlett-kernel HAC standard errors. A companion VAR in $Y_t = [\Delta Issuance_t$, Δy^{1M}_t, Δy^{3M}_t, Δy^{10Y}_t, $\Delta VIX_t]'$ with AIC-selected lags and Cholesky ordering (issuance $\rightarrow$ short yields $\rightarrow$ long yields $\rightarrow$ VIX) provides a dynamic cross-check.

由于收益率会反馈到发行激励——更高的国库券收益率提高储备收益并吸引供给——流量变量是内生的。本文借鉴 **(author?)** [1]，构造 Elastic Net 残差工具变量：将加密原生变量对传统金融信息集做交叉验证的弹性网络回归并保留残差，从而分离出与债市状况正交的加密特有波动。具体地，IV1 为比特币对数收益对 $\Delta \ln SP500$、$\Delta \ln DXY$、$\Delta \ln Gold$、ΔVIX、ΔDFF 与 Δy^{10Y} 回归的残差；IV2 为 DeFi TVL 的同类残差；IV3 为去均

值的 USDT 场外溢价。第一阶段为 $Flow_t^{(5d)} = \pi_0 + \pi_1 Z_t + \Pi X_t + u_t$，第二阶段用 $\widehat{Flow}$ 替换式 (1) 中的 $Flow$，采用 Bartlett 核 HAC 标准误的 2SLS 估计。作为动态交叉验证，另估计 $Y_t = [\Delta Issuance_t, \Delta y_t^{1M}, \Delta y_t^{3M}, \Delta y_t^{10Y}, \Delta VIX_t]'$ 的 VAR（AIC 定阶，Cholesky 排序：净发行 → 短端收益率 → 长端收益率 → VIX）。

4.2. Spillover measurement / 溢出测度

For Q2 we estimate a VAR(2) in $Y_t = [\Delta MC_t, \Delta y_t^{3M}, \Delta y_t^{2Y}, \Delta y_t^{10Y}, \Delta VIX_t, \Delta MM_t]'$, where ΔMC is the winsorized stablecoin market-cap growth and ΔMM is the change in the SOFR–ON-RRP spread (the full-sample credit/liquidity proxy, since the OAS series start only in 2023). From the generalized forecast-error variance decomposition at horizon $H = 10$ we compute the **(author?)** [4] total connectedness index

$$S(H) = \frac{\sum_{i \neq j} \tilde{\theta}_{ij}^{g}(H)}{N} \times 100, \tag{2}$$

together with directional TO/FROM spillovers and net spillovers $S_i^{net} = S_{\bullet \leftarrow i} - S_{i \leftarrow \bullet}$. Time variation is captured by re-estimating the system in rolling 250-trading-day windows, a standard approximation to the TVP-VAR connectedness of **(author?)** [5].

针对 Q2，本文估计 $Y_t = [\Delta MC_t, \Delta y_t^{3M}, \Delta y_t^{2Y}, \Delta y_t^{10Y}, \Delta VIX_t, \Delta MM_t]'$ 的 VAR(2)，其中 ΔMC 为缩尾的稳定币市值增长率，ΔMM 为 SOFR–ON-RRP 利差变动（因 OAS 序列 2023 年才开始，以其作为全样本信用/流动性代理）。由 horizon $H = 10$ 的广义预测误差方差分解计算 **(author?)** [4] 总连通性指数（式 (2)），以及方向性 TO/FROM 溢出与净溢出 $S_i^{net} = S_{\bullet \leftarrow i} - S_{i \leftarrow \bullet}$。时变特征通过在 250 个交易日的滚动窗口内重估该系统捕捉，这是 **(author?)** [5] 的 TVP-VAR 连通性的标准滚动近似。

4.3. Credit-spread regressions and event studies / 信用利差回归与事件研究

For Q3 the estimating equation is

$$\Delta CreditSpread_t = \alpha_2 + \beta_2 \cdot \widehat{Flow}_t^{(5d)} + \gamma_2 X_t + \varepsilon_{2t}, \tag{3}$$

estimated by OLS and 2SLS (IV1 and IV3) for four spread proxies on their respective available samples: investment-grade OAS (2023-06 onward), high-yield OAS (2023-06 onward), the TED spread (2020-01 to 2022-01) and the SOFR–ON-RRP spread (full sample). The event study follows **(author?)**

[6]: for each event e and window $[\tau_1, \tau_2]$ (with $[-5, +5]$ and $[-10, +10]$ trading days),

$$CAR_{[\tau_1,\tau_2]} = \sum_{t=\tau_1}^{\tau_2} \left(\Delta Spread_t - \mathbb{E}[\Delta Spread_t]\right), \tag{4}$$

where the expectation is the mean over the 60 trading days ending six days before the event, and inference uses the estimation-window standard deviation. The event library covers the Tether commercial-paper disclosure scrutiny (2021-10-15), the Terra/UST collapse (2022-05-09), Tether's zero-CP commitment (2022-07-27), the USDC depeg (2023-03-10) and the GENIUS Act Senate passage (2025-06-17).

针对 Q3，估计方程为式 (3)，分别用 OLS 与 2SLS（IV1 与 IV3）对四个利差代理在各自可用样本上估计：投资级 OAS（2023-06 起）、高收益 OAS（2023-06 起）、TED 利差（2020-01 至 2022-01）与 SOFR–ON-RRP 利差（全样本）。事件研究遵循 **(author?)** [6]：对每个事件 e 与窗口 $[\tau_1, \tau_2]$（$[-5, +5]$ 与 $[-10, +10]$ 个交易日），按式 (4) 计算累计异常变动，其中期望取事件前第 6 个交易日往前 60 个交易日的均值，推断基于估计窗标准差。事件库涵盖 Tether 商业票据披露审查（2021-10-15）、Terra/UST 崩盘（2022-05-09）、Tether 零 CP 承诺（2022-07-27）、USDC 脱锚（2023-03-10）与 GENIUS Act 参议院通过（2025-06-17）。

4.4. Heteroskedasticity identification and stress tests / 异方差识别与压力测试

For Q4, **(author?)** [7] identification exploits the covariance shift between normal and crisis regimes. Under the assumption that the variance of the stablecoin shock rises in crisis windows while other shock variances are unchanged, the coefficient of stablecoin shocks on bond-market variables is

$$\hat{\beta}_R = \frac{Cov_{crisis}(\Delta MC, \Delta y) - Cov_{normal}(\Delta MC, \Delta y)}{Var_{crisis}(\Delta MC) - Var_{normal}(\Delta MC)}, \tag{5}$$

with crisis windows defined as the UST collapse (2022-05-09 to 2022-05-31), the FTX episode (2022-11-08 to 2022-11-14) and the USDC depeg (2023-03-10 to 2023-03-24). The heteroskedasticity premise is tested with a Brown–Forsythe variance-ratio test, and confidence intervals come from 1,000 bootstrap replications.

针对 Q4，**(author?)** [7] 识别利用正常区制与危机区制之间的协方差移动。在危机窗口内稳定币冲击方差上升而其他冲击方差不变的假设下，稳定币冲击对债市变量的系数由式 (5) 给出，危机窗口定义为 UST 崩

盘（2022-05-09 至 2022-05-31）、FTX 事件（2022-11-08 至 2022-11-14）与 USDC 脱锚（2023-03-10 至 2023-03-24）。异方差前提用 Brown–Forsythe 方差比检验，置信区间来自 1,000 次 bootstrap。

Stress tests extrapolate the estimated impulse responses to four scenarios: (S1) a 10% USDT redemption, i.e. roughly \$18.6 bn of forced sales at the current market size; (S2) an SVB-like USDC depeg, calibrated to the observed \$4.1 bn USDC contraction of 2023-03-10 to 2023-03-14; (S3) a \$50 bn joint run on major stablecoins; (S4) a \$30 bn regulatory rebalancing out of credit and risky assets. Impacts are computed as the scenario size times the peak LP coefficient for the 1M, 3M and 10Y tenors, with the asymmetric outflow coefficient providing an upper bound for run scenarios, and the 2SLS spread coefficients mapping the same shocks into IG/HY spread changes. All stress figures carry an explicit linear-extrapolation caveat.

压力测试将估计的脉冲响应外推到四个情景：（S1）10% 的 USDT 赎回，按当前规模约 186 亿美元的被迫抛售；（S2）类 SVB 的 USDC 脱锚，校准到 2023-03-10 至 2023-03-14 实际观测到的 41 亿美元 USDC 收缩；（S3）主要稳定币 500 亿美元联合挤兑；（S4）300 亿美元监管强制从信用与风险资产再平衡。冲击影响等于情景规模乘以 1M、3M 与 10Y 期限的 LP 峰值系数，非对称流出系数为挤兑情景提供上限，2SLS 利差系数将同一冲击映射为 IG/HY 利差变化。所有压力测试数字均附带明确的线性外推注意事项。

4.5. Interpretable machine learning / 可解释机器学习

As a nonlinear cross-check, an XGBoost regressor [14] predicts the five-day-ahead cumulative change in the 3-month yield from the full daily information set (stablecoin flows, on-chain activity, derivatives, macro-financial controls), trained on the first 80% of the sample and tested on the last 20%. SHAP values [15] decompose predictions into feature contributions, allowing us to verify whether stablecoin flows carry incremental information beyond standard risk factors.

作为非线性交叉验证，本文用 XGBoost 回归器 [14] 以全部日度信息集（稳定币流量、链上活跃度、衍生品、宏观金融控制）预测未来 5 日 3 个月期收益率的累计变动，前 80% 样本训练、后 20% 样本测试。SHAP 值 [15] 将预测分解为各特征贡献，用以检验稳定币流量是否包含超越标准风险因子的增量信息。

5. Empirical Results / 实证结果

5.1. Q1: Stablecoin flows and the Treasury yield curve / Q1：稳定币流量与国债收益率曲线

Table 2 and Figure 2 report the local-projection estimates of equation (1). Three regularities stand out. First, the impact response at the very front of the curve is negative: a \$1 bn five-day net issuance lowers the 1-month yield by 0.23 bps on impact ($p = 0.043$). Second, the effect builds up over roughly three weeks: at $h = 16$ the 1-month response reaches -3.40 bps ($p = 0.011$) and remains at -2.95 bps at $h = 20$ ($p = 0.030$); the 3-month response is significant from $h = 4$ (-0.97 bps, $p = 0.001$) through $h = 20$ (-2.40 bps, $p = 0.020$). Third, the effect is economically and statistically confined to the short end: coefficients at the 1Y tenor are small and insignificant, and at 2Y–30Y they are positive but never significant at conventional levels. This term-structure profile is precisely what a preferred-habitat reading of stablecoin reserve portfolios predicts, supporting H1a and H1d.

表 2 与图 2 报告式 (1) 的局部投影估计。三个规律尤为突出。第一，曲线最前端的即期响应为负：10 亿美元的 5 日净发行在当期使 1 个月期收益率下行 0.23 个基点（$p = 0.043$）。第二，效应在约三周内逐步累积：$h = 16$ 时 1 个月期响应达 -3.40 个基点（$p = 0.011$），$h = 20$ 时仍为 -2.95 个基点（$p = 0.030$）；3 个月期响应从 $h = 4$（-0.97 个基点，$p = 0.001$）到 $h = 20$ （-2.40 个基点，$p = 0.020$）持续显著。第三，效应在经济与统计意义上均局限于短端：1Y 期限系数很小且不显著，2Y–30Y 期限系数为正但在常规水平上从不显著。这一期限结构形态恰是稳定币储备组合的偏好栖息地解读所预测的，支持 H1a 与 H1d。

The instrumental-variable stage delivers mixed evidence. The first stage is strong for the DeFi-TVL residual instrument ($F = 13.2$) but weak for the Bitcoin residual ($F = 0.5$) and the OTC premium ($F = 1.2$), so we headline the IV2 specification (Figure 3). Second-stage point estimates are imprecise and predominantly positive at all tenors (e.g. +4.1 bps, $p = 0.43$, at the 3M tenor and $h = 10$), which we do not interpret as evidence against the LP result: with weak-to-moderate instruments the 2SLS confidence intervals are wide and include the LP estimates, and the exclusion restriction is harder to defend for TVL than for bill yields. We therefore regard the IV estimates as imprecise bounds and the LP estimates, whose significance survives the robustness battery of Section 6, as the preferred reduced-form evidence of a causal channel.

Table 2: 局部投影基准结果（β_h^τ，每 10 亿美元 5 日净流量的基点响应）/ LP baseline results (β_h^τ, bp per \$1 bn five-day net flow)

Tenor 期限	$h = 0$	$h = 5$	$h = 10$	$h = 20$
1M	-0.229^{**} (0.113)	-0.408 (0.326)	-1.069 (0.797)	-2.949^{**} (1.363)
3M	0.096 (0.060)	-0.820^{***} (0.277)	-1.186^{**} (0.576)	-2.401^{**} (1.031)
1Y	0.065 (0.104)	-0.165 (0.344)	-0.159 (0.626)	-0.877 (1.039)
2Y	0.203 (0.124)	0.172 (0.404)	0.376 (0.751)	0.398 (1.086)
5Y	0.221^{*} (0.125)	0.165 (0.374)	0.611 (0.736)	0.790 (1.067)
10Y	0.175 (0.118)	0.082 (0.341)	0.453 (0.691)	0.439 (1.025)
30Y	0.172 (0.115)	0.017 (0.324)	0.224 (0.596)	0.195 (0.916)

注：括号内为 Newey–West HAC 标准误（滞后 $h+2$）；*、**、*** 分别表示 10%、5%、1% 显著性。控制变量含 ΔDFF、$\Delta SOFR$、ΔVIX、$\Delta \ln DXY$、$\Delta \ln SP500$。Notes: Newey–West HAC standard errors (lag $h+2$) in parentheses; *, **, *** denote 10%, 5%, 1% significance. Controls: ΔDFF, $\Delta SOFR$, ΔVIX, $\Delta \ln DXY$, $\Delta \ln SP500$.

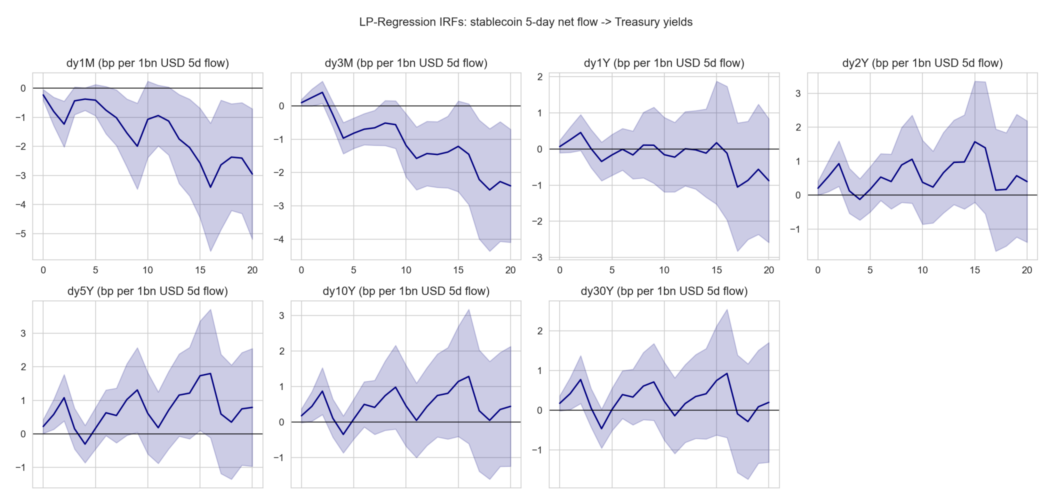

Figure 2: LP 基准脉冲响应：稳定币 5 日净流量对七个期限国债收益率的影响（阴影为 90% 置信区间）/ LP baseline IRFs of seven Treasury tenors to the five-day stablecoin net flow (shaded: 90% confidence bands)

工具变量估计给出了混合证据。第一阶段对 DeFi-TVL 残差工具较强（$F = 13.2$），但对比特币残差（$F = 0.5$）与场外溢价（$F = 1.2$）较弱，因此正文以 IV2 规格为主（图 3）。第二阶段点估计在所有期限上均不精确且以正值为主（如 3M 期限 $h = 10$ 处为 +4.1 个基点，$p = 0.43$），本文不将其解读为反驳 LP 结果的证据：在工具变量偏弱至中等的条件下，2SLS 置信区间很宽且包含 LP 估计值，且 TVL 的排他性约束比国库券收益率更难辩护。因此本文将 IV 估计视为不精确的区间参考，而以经受住第 6 节稳健性检验的 LP 估计作为因果渠道的优选约简式证据。

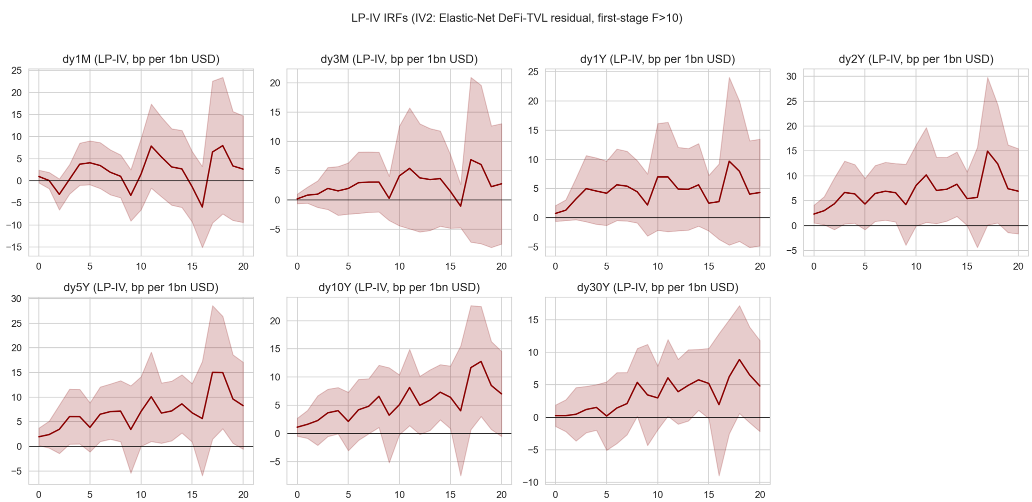

Figure 3: LP-IV 脉冲响应（IV2：Elastic Net 的 DeFi-TVL 残差，第一阶段 $F = 13.2$）/ LP-IV IRFs (IV2: Elastic-Net DeFi-TVL residual, first-stage $F = 13.2$)

The VAR cross-check (Figure 4) is consistent with the LP evidence. A one-unit ($1 bn) Cholesky issuance shock lowers the 1-month yield by 0.26 bps one day after impact, while the 3-month and 10-year responses are small and short-lived; the VIX response is negligible after a few days. The AIC selects four lags. Taken together, the reduced-form evidence supports H1a and H1d: stablecoin net issuance compresses the very front of the curve with a delay of one to three weeks, and the effect does not travel up the curve.

VAR 交叉验证（图 4）与 LP 证据一致。一单位（10 亿美元）Cholesky 发行冲击在冲击后一日使 1 个月期收益率下行 0.26 个基点，3 个月期与 10 年期响应很小且转瞬即逝，VIX 响应在数日后可忽略。AIC 选择 4 阶滞后。综合来看，约简式证据支持 H1a 与 H1d：稳定币净发行在一至三周的时滞后压缩曲线最前端，且效应不向长端传导。

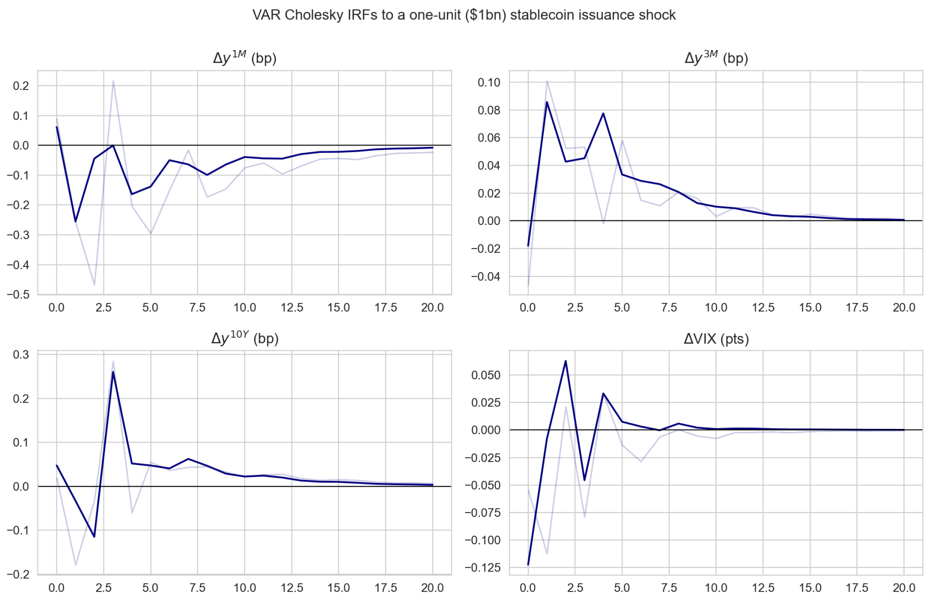

Figure 4: VAR Cholesky 脉冲响应：一单位（10 亿美元）稳定币净发行冲击 / VAR Cholesky IRFs to a one-unit ($1 bn) stablecoin issuance shock

5.2. Q2: Volatility spillovers / *Q2*：波动溢出

The full-sample Diebold–Yilmaz decomposition (Table 3 and Figure 5) yields a total connectedness index of 19.7%: about one fifth of the ten-day-ahead forecast-error variance in the six-variable system is cross-variable. The stablecoin market is close to neutral on average, with a net spillover of −0.14 pp: it transmits about as much variance as it receives. The 2-year and 10-year yields are the system's hubs (net +6.6 pp and +0.9 pp), the 3-month yield and the money-market stress proxy are net receivers (−5.2 pp and −6.4 pp), and VIX is a moderate net transmitter (+4.3 pp). H2a is therefore supported: connectedness is clearly positive, though the system is dominated by within-block variance.

全样本 Diebold–Yilmaz 分解（表 3 与图 5）给出 19.7% 的总连通性指数：六变量系统中约五分之一的十日前瞻预测误差方差来自跨变量溢出。稳定币市场平均接近中性，净溢出为 −0.14 个百分点：其发出的方差份额与接收的大致相当。2 年期与 10 年期收益率是系统枢纽（净 +6.6 与 +0.9 个百分点），3 个月期收益率与货币市场压力代理是净接收者（−5.2 与 −6.4 个百分点），VIX 为中等净输出者（+4.3 个百分点）。因此 H2a 得到支持：连通性明确为正，尽管系统由组内方差主导。

Table 3: Diebold–Yilmaz 方向性溢出（%，$H = 10$，VAR(2)）/ Diebold–Yilmaz directional spillovers (%, $H = 10$, VAR(2))

Variable 变量	TO 发出	FROM 接收	NET 净溢出
ΔMC（稳定币市值增长）	1.14	1.28	−0.14
Δy^{3M}	15.45	20.69	−5.24
Δy^{2Y}	49.94	43.29	+6.65
Δy^{10Y}	41.10	40.23	+0.86
ΔVIX	8.20	3.91	+4.29
ΔMM（SOFR–RRP 变动）	2.25	8.68	−6.42
TCI 总溢出指数 = 19.68%			

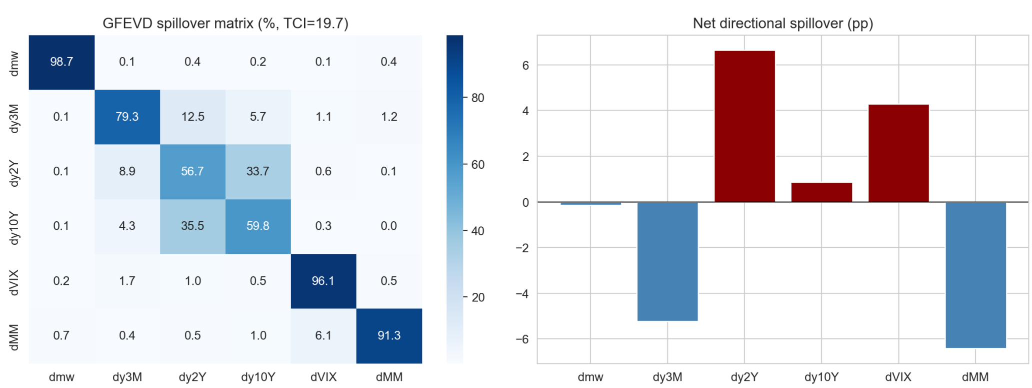

Figure 5: DY 溢出矩阵（左）与净方向性溢出（右）/ DY spillover matrix (left) and net directional spillovers (right)

The rolling time-varying estimates (Figure 6) qualify the average picture in two ways. First, connectedness is strongly time-varying: the rolling TCI peaks at 32% in the COVID turmoil of 2020, bottoms near 12–13% in early 2021, and trends upward from 2023 onward, stabilizing around 26% in 2024–2026—the stablecoin–bond nexus has strengthened as the sector has grown. Second, the sign of the stablecoin net spillover flips in stress: it averages +1.7 pp in the crisis windows against −2.4 pp elsewhere, and reaches its maximum of +5.8 pp on 2023-03-22, days after the USDC depeg. H2c is thus supported, while H2b receives only qualified support: the event-window TCI average (22.4%) is slightly below the rest-of-sample mean (22.8%) because the rolling window dilutes short-lived crisis spikes.

滚动时变估计（图 6）从两方面修正了全样本平均图景。第一，连通性高度时变：滚动 TCI 在 2020 年 COVID 动荡中达 32% 的峰值，2021 年初降至 12–13% 附近，2023 年后趋势性上行，2024–2026 年稳定在 26% 左右——随着行业规模扩大，稳定币与债市的联动增强。第二，稳定币净溢出的符号在压力时期反转：危机窗口内均值为 +1.7 个百分点，其余时期为 −2.4 个百分点，并在 2023-03-22（USDC 脱锚数日后）达到 +5.8 个百分点的最大值。因此 H2c 得到支持，而 H2b 仅获有限支持：事件窗口 TCI 均值（22.4%）略低于样本其余部分均值（22.8%），原因是滚动窗口稀释了短暂的危机尖峰。

5.3. Q3: Credit spreads and event evidence / Q3：信用利差与事件证据

Regression-based evidence on the credit-spread channel is weak (Figure 7). OLS and 2SLS estimates of equation (3) are statistically insignificant for all four spread proxies, although the OLS point estimates are uniformly negative (e.g. −0.16 bps per $1 bn for the TED spread, −0.04 bps for the SOFR–RRP spread), consistent in sign with H3a. We read this as evidence that daily spread variation is dominated by aggregate risk factors, so that any stablecoin-flow component is too small to isolate at daily frequency with available proxies—not as evidence of absence, given that the instruments are weak in these subsamples (first-stage F between 0.5 and 4.9).

信用利差渠道的回归证据较弱（图 7）。式 (3) 的 OLS 与 2SLS 估计在四个利差代理上均不显著，尽管 OLS 点估计一致为负（如 TED 利差每 10 亿美元 −0.16 个基点，SOFR–RRP 利差 −0.04 个基点），符号上与 H3a 一致。本文将其解读为日度利差变动由总体风险因子主导、以致在现有代理下无法从日度频率分离出稳定币流量分量的证据，而非" 无效应" 的证据——因为在这些子样本中工具变量较弱（第一阶段 F 介于 0.5 至 4.9）。

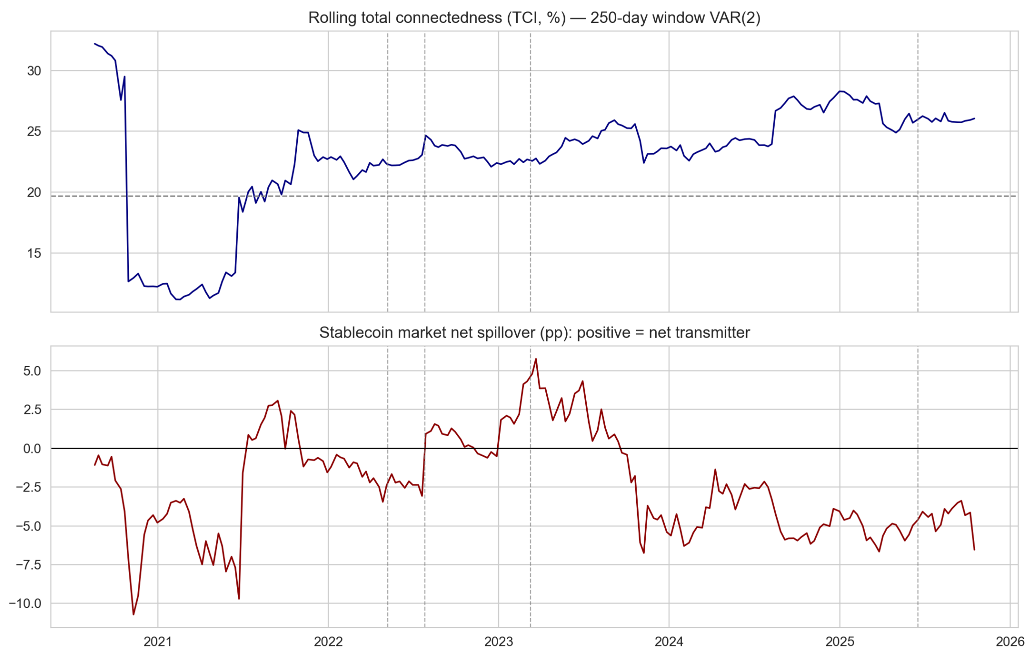

Figure 6: 滚动时变连通性：总溢出指数（上）与稳定币净溢出（下，正值 = 净输出）/ Rolling time-varying connectedness: total spillover index (top) and stablecoin net spillover (bottom, positive = net transmitter)

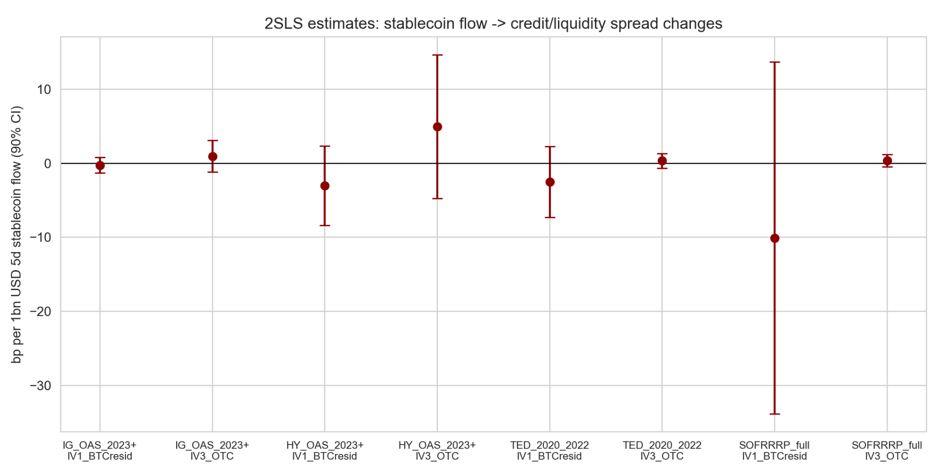

Figure 7: 2SLS 估计：稳定币流量对信用/流动性利差变动的影响（90% 置信区间）/ 2SLS estimates of stablecoin flows on credit/liquidity spread changes (90% confidence intervals)

Event-time evidence is sharper (Table 4 and Figure 8). The USDC depeg is the clearest case: within five trading days the SOFR–RRP spread widened by an abnormal +3 bps ($p = 0.045$) and VIX rose by 7.4 points ($p = 0.001$), while the 3-month bill yield fell by an abnormal 32 bps over ten days ($p = 0.001$)—a classic flight-to-quality pattern in which run risk inside the stablecoin sector spills into money-market stress while bills rally on safe-haven demand. Tether's zero-CP commitment is associated with an abnormal −22.4 bps five-day move in the 3-month yield ($p = 0.085$), and the GENIUS Act Senate passage with a +7.9-point VIX move ($p = 0.020$) and a borderline +8.6 bps widening of investment-grade OAS over ten days ($p = 0.105$). The event evidence thus supports H3b with an important qualification: regulatory rebalancing away from riskier reserve assets coincides with wider credit spreads and lower bill yields, not wider bill yields, because the safe-asset leg dominates in the window.

事件时间维度的证据更为清晰（表 4 与图 8）。USDC 脱锚是最明确的案例：五个交易日内 SOFR–RRP 利差异常走阔 +3 个基点（$p = 0.045$），VIX 上升 7.4 个点（$p = 0.001$），而 3 个月期国库券收益率十日内异常下行 32 个基点（$p = 0.001$）——这是典型的避险模式：稳定币部门内部的挤兑风险外溢为货币市场压力，而国库券因避险需求上涨。Tether 零 CP 承诺对应 3 个月期收益率五日 −22.4 个基点的异常变动（$p = 0.085$），GENIUS Act 参议院通过对应 VIX +7.9 个点的变动（$p = 0.020$）以及投资级 OAS 十日 +8.6 个基点的边际走阔（$p = 0.105$）。因此事件证据支持 H3b，但附重要限定：迫使储备减持风险资产的监管再平衡伴随信用利差走阔与国库券收益率下行（而非上行），因为在事件窗口内安全资产腿占主导。

5.4. Q4: Heteroskedasticity identification and stress tests / *Q4*：异方差识别与压力测试

The heteroskedasticity premise is partially supported: the variance of stablecoin market-cap growth is 1.27 times higher in crisis windows than in normal times (Brown–Forsythe $p = 0.086$ for dy3M pairings; $p = 0.020$ for the money-market pairing). The Rigobon point estimates are, however, imprecise: −19.6 bps per percentage point of market-cap change for the 3-month yield with a 90% bootstrap interval of $[-111, 72]$, and −0.18 bps for the money-market spread with interval $[-4.9, 5.1]$. With 24–45 crisis observations the covariance difference is noisy, so H4a receives only weak support; Figure 9 visualizes the regime shift that the estimator exploits.

异方差前提获得部分支持：危机窗口内稳定币市值增长率的方差是正常时期的 1.27 倍（dy3M 配对的 Brown–Forsythe 检验 $p = 0.086$；货币市场

Table 4: 事件研究：累计异常变动（CAR）摘要 / Event-study CAR summary

Event 事件	Variable 变量	Window 窗口	CAR	p
USDC depeg (2023-03-10)	ΔSOFR–RRP (bp)	$[-5, +5]$	+3.0	0.045
USDC depeg	ΔVIX (pts)	$[-5, +5]$	+7.4	0.001
USDC depeg	Δy^{3M} (bp)	$[-10, +10]$	−32.1	0.001
Tether zero-CP (2022-07-27)	Δy^{3M} (bp)	$[-5, +5]$	−22.4	0.085
Tether zero-CP	Δy^{3M} (bp)	$[-10, +10]$	−29.8	0.139
GENIUS Senate (2025-06-17)	ΔVIX (pts)	$[-5, +5]$	+7.9	0.020
GENIUS Senate	ΔIG OAS (bp)	$[-10, +10]$	+8.6	0.105
GENIUS Senate	ΔHY OAS (bp)	$[-10, +10]$	+30.6	0.364

注：预期变动取事件前第 6 个交易日往前 60 个交易日的均值；t 统计量基于估计窗标准差。完整 36 个事件-变量-窗口组合见 `tables/cp_event_study_results.csv`。Notes: expected changes are means over the 60 trading days ending six days before the event; t-statistics use the estimation-window standard deviation. All 36 event–variable–window combinations are in `tables/cp_event_study_results.csv`.

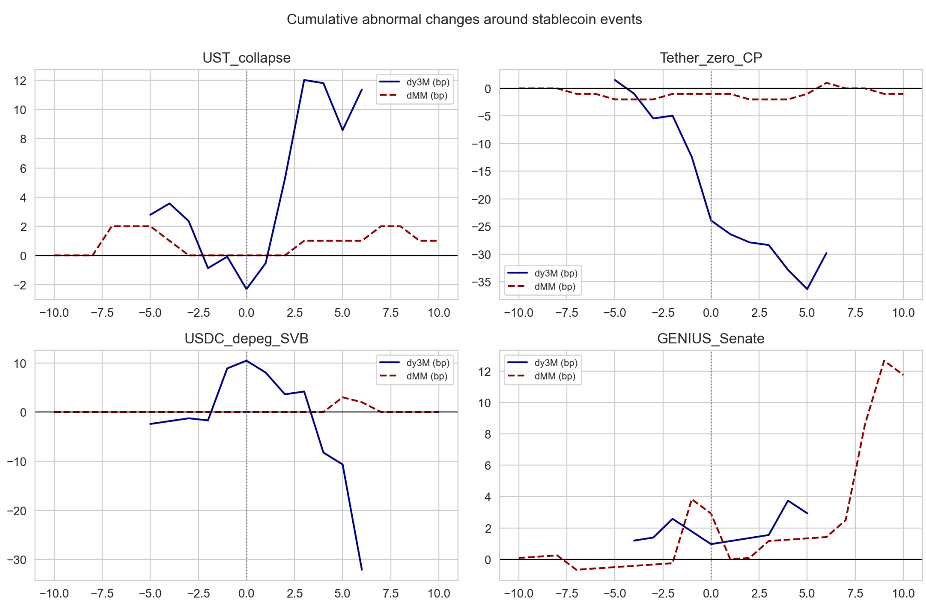

Figure 8: 稳定币事件前后 Δy^{3M} 与 ΔMM 的累计异常变动 / Cumulative abnormal changes in Δy^{3M} and ΔMM around stablecoin events

配对 $p = 0.020$）。然而 Rigobon 点估计不精确：3 个月期收益率对市值变动每百分点的系数为 -19.6 个基点，90% bootstrap 区间为 $[-111, 72]$；货币市场利差的系数为 -0.18 个基点，区间为 $[-4.9, 5.1]$。危机观测仅 24–45 个，协方差差异噪声较大，因此 H4a 仅获弱支持；图 9 可视化了该估计量所利用的区制移动。

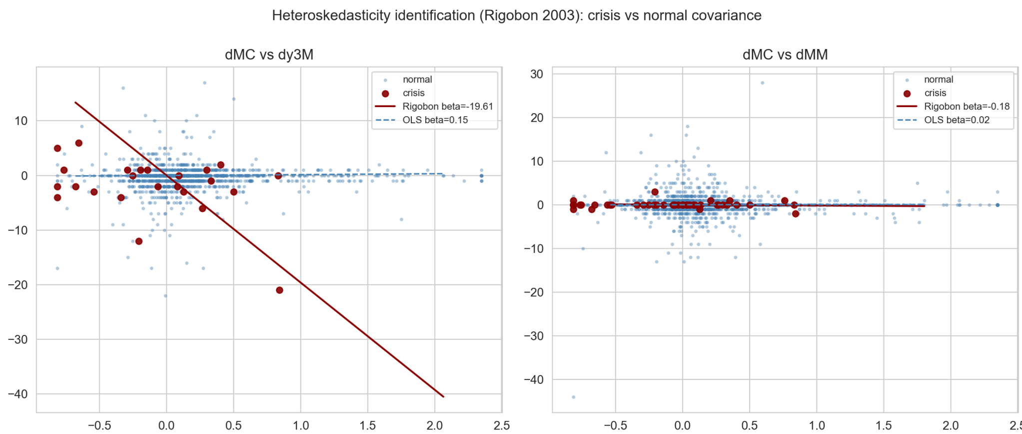

Figure 9: 异方差识别：危机期与正常期的协方差结构差异 / Heteroskedasticity identification: covariance shift between crisis and normal regimes

The calibrated stress tests (Table 5 and Figure 10) translate the estimates into systemic-risk magnitudes. A 10% USDT run ($18.6 bn of forced sales) raises the 1-month yield by 63 bps and the 3-month yield by 47 bps under the symmetric LP calibration, or 60 bps under the asymmetric outflow calibration, while high-yield spreads widen by about 57 bps. The SVB-like USDC scenario ($4.1 bn) implies a 10–13 bps rise in the 3-month yield—broadly consistent with the opposite (flight-to-quality) sign observed in the event study, which underscores that the 2023 episode combined a stablecoin run with an aggregate banking shock whose bill-rallying force dominated. A $50 bn joint run raises the 3-month yield by 126–161 bps and high-yield spreads by 152 bps; the regulatory rebalancing scenario ($30 bn out of risky assets) widens high-yield spreads by 91 bps. These figures assume linear extrapolation of daily-frequency coefficients over multi-week horizons and should be read as upper-bound orders of magnitude, not point forecasts.

校准压力测试（表 5 与图 10）将估计值转化为系统性风险量级。10% 的 USDT 挤兑（186 亿美元被迫抛售）在对称 LP 校准下推高 1 个月期收益率 63 个基点、3 个月期收益率 47 个基点，在非对称流出校准下为 60 个基点，

同时高收益利差走阔约 57 个基点。类 SVB 的 USDC 情景（41 亿美元）对应 3 个月期收益率上行 10–13 个基点——与事件研究中观测到的相反（避险）符号大致相容，这凸显 2023 年事件是稳定币挤兑与总体银行业冲击的叠加，后者压低国库券收益率的力量占主导。500 亿美元联合挤兑推高 3 个月期收益率 126–161 个基点、高收益利差 152 个基点；监管再平衡情景（300 亿美元撤离风险资产）使高收益利差走阔 91 个基点。这些数字假设日度频率系数在多周 horizon 上的线性外推，应视为上限量级而非点预测。

Table 5: 压力测试情景：隐含债市冲击（bp，线性外推）/ Stress-test scenarios: implied bond-market impacts (bp, linear extrapolation)

Scenario 情景	Size 规模 ($bn)	Δy^{1M}	Δy^{3M} 中心	Δy^{3M} 非对称上限	Δy^{10Y}	ΔHY OAS
S1: USDT 10% 赎回	−18.6	+63.4	+47.1	+60.2	−23.9	+56.6
S2: USDC 脱锚（类 SVB）	−4.1	+13.9	+10.3	+13.2	−5.2	+12.4
S3: 500 亿联合挤兑	−50.0	+170.1	+126.2	+161.3	−64.1	+151.9
S4: 监管再平衡 300 亿	−30.0	+102.1	+75.7	+96.8	−38.4	+91.1

注：中心列为 LP 峰值系数校准；” 非对称上限” 为流出方向系数校准；HY OAS 列用 2SLS 系数。所有数字受线性外推限制。Notes: central columns use peak LP coefficients; the asymmetric upper bound uses the outflow coefficient; the HY OAS column uses the 2SLS coefficient. All figures are subject to linear-extrapolation caveats.

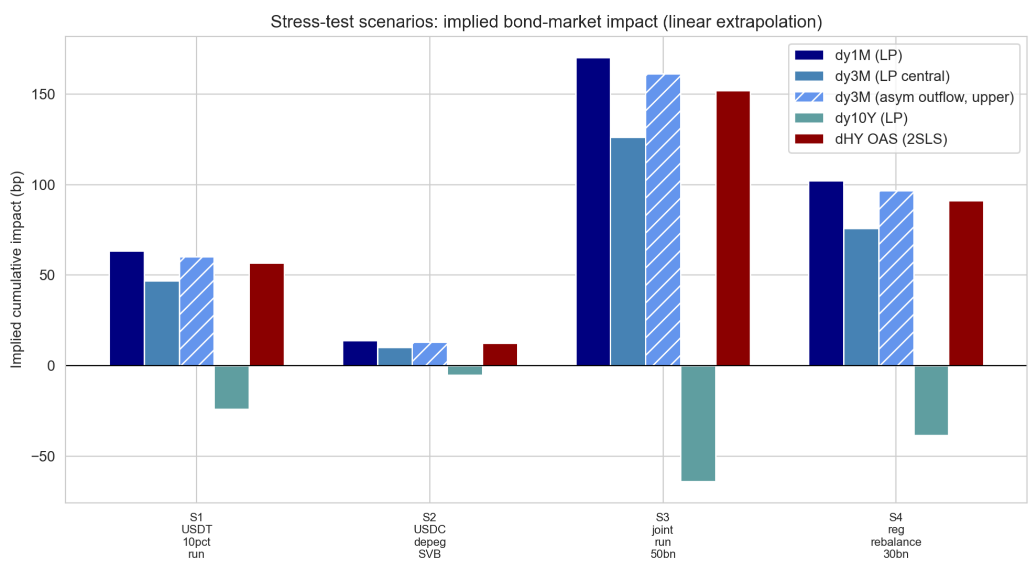

Figure 10: 压力测试：四情景下的隐含累计冲击 / Stress tests: implied cumulative impacts under four scenarios

6. Heterogeneity and Robustness / 异质性与稳健性

6.1. Asymmetry and issuer heterogeneity / 非对称性与发行人异质性

Splitting the five-day flow into its positive (issuance) and negative (redemption) parts confirms the hypothesized asymmetry in magnitude: at $h = 10$ the 3-month yield responds by -0.49 bps per \$1 bn of inflows but -3.23 bps per \$1 bn of outflows (Figure 11). The difference is not statistically significant (Wald $p = 0.178$), so H1c is supported in magnitude but not in statistical precision at daily frequency. Issuer-level estimates reveal that the USDC flow drives the headline result (-2.19 bps per \$1 bn, $p = 0.011$ at $h = 10$) while the USDT coefficient is smaller and insignificant (-0.83 bps, $p = 0.36$)—plausible given that USDC redemptions are operationally concentrated in U.S. banking hours and settle directly against Treasury and repo collateral. The crisis-only subsample is too small ($n = 9$) for inference; its positive point estimate is consistent with redemption pressure dominating in stress but is imprecise.

将 5 日流量拆分为正部（发行）与负部（赎回）后，假说的量级非对称性得到确认：$h = 10$ 处 3 个月期收益率对每 10 亿美元流入的响应为 -0.49 个基点，对每 10 亿美元流出为 -3.23 个基点（图 11）。两者差异在统计上不显著（Wald $p = 0.178$），因此 H1c 在日度频率上获得量级支持而非统计精度支持。发行人层面估计显示，USDC 流量主导了总体结果（每 10 亿美元 -2.19 个基点，$h = 10$ 处 $p = 0.011$），而 USDT 系数较小且不显著（-0.83 个基点，$p = 0.36$）——考虑到 USDC 赎回在操作上集中于美国银行营业时段并直接以国债与回购抵押品结算，这一差异是合理的。危机子样本太小（$n = 9$）无法做推断；其正的点估计与压力期赎回压力主导一致，但不精确。

6.2. Robustness battery / 稳健性检验

Table 6 collects the robustness battery for the headline specification (3-month yield, $h = 10$). The negative sign survives every perturbation of the flow definition and sample: the one-day flow (-2.25, $p = 0.139$), the winsorized flow (-1.20, $p = 0.047$), the crisis-excluded sample (-1.29, $p = 0.047$) and the weekly aggregation (-2.45, $p = 0.019$) all preserve or strengthen the baseline estimate of -1.19 bps. The LP-IV variant with the strong-instrument TVL residual is insignificant, as discussed in Section 5.1. The machine-learning cross-check confirms that stablecoin flows carry incremental information: in the XGBoost SHAP decomposition (Figure 12), the five-day flow ranks third among seventeen predictors, behind the Fear & Greed

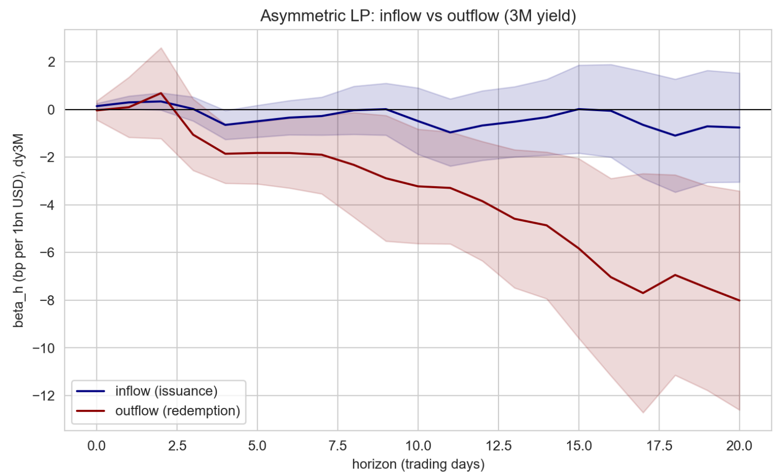

Figure 11: 非对称局部投影：流入 vs 流出（3M 收益率）/ Asymmetric local projections: inflow vs outflow (3M yield)

Index and futures leverage and ahead of VIX changes and equity returns. Out-of-sample predictability of five-day yield changes is nevertheless poor ($R^2_{OS} = -4.57$), consistent with the near-random-walk nature of daily bill yields and reinforcing the choice of linear projections for inference.

表 6 汇总了针对基准规格（3 个月期收益率，$h = 10$）的稳健性检验。负号在流量定义与样本的所有扰动下均成立：1 日流量（-2.25，$p = 0.139$）、缩尾流量（-1.20，$p = 0.047$）、剔除危机样本（-1.29，$p = 0.047$）与周度聚合（-2.45，$p = 0.019$）均保持或强化了 -1.19 个基点的基准估计。采用强工具 TVL 残差的 LP-IV 变体不显著，如第 5.1 节所讨论。机器学习交叉验证确认稳定币流量包含增量信息：在 XGBoost 的 SHAP 分解中（图 12），5 日流量在 17 个预测变量中排第三，次于恐惧贪婪指数与期货杠杆率，先于 VIX 变动与股票收益。不过，5 日收益率变动的样本外可预测性仍然很差（$R^2_{OS} = -4.57$），这与日度国库券收益率近似随机游走的性质一致，并强化了选择线性投影做统计推断的合理性。

7. Economic Implications and Policy Discussion / 经济含义与政策讨论

The estimates carry three economic messages. First, the yield-compression footprint of stablecoins is real but modest per unit of flow and concentrated at the bill tenors where reserves are invested; aggregated over the sector's

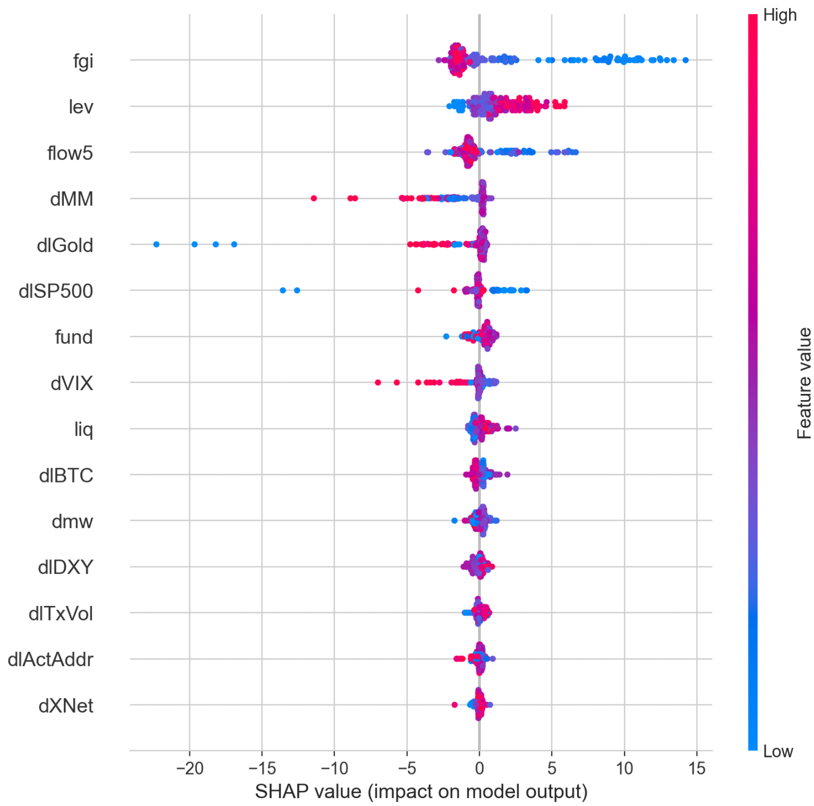

Figure 12: XGBoost 的 SHAP 特征贡献分解 / SHAP feature-contribution decomposition of the XGBoost model

Table 6: 稳健性检验（被解释变量：Δy^{3M}，$h=10$）/ Robustness battery (dependent variable: Δy^{3M}, $h=10$)

Specification 规格	β (bp/\$1bn)	SE	n
Baseline 基准（5 日流量）	-1.186^{**}	0.576	548
R1: 1 日流量	-2.245	1.515	549
R4: 缩尾流量	-1.204^{**}	0.607	548
R3: 剔除危机窗口	-1.286^{**}	0.648	539
R5: 周度聚合（h=2 周）	-2.446^{**}	1.046	320
R2: LP-IV（TVL 残差 IV）	4.084	5.174	548
USDT 流量	-0.826	0.902	548
USDC 流量	-2.186^{**}	0.855	548
流入（正部）	-0.490	0.846	548
流出（负部）	-3.226^{**}	1.458	548

注：** 表示 5% 显著。R2 规格的第一阶段 $F=7.6$（该 horizon 下）。Notes: ** denotes 5% significance. The R2 specification has a first-stage F of 7.6 at this horizon.

growth, however, the cumulative contribution to short-end demand is material: the \$258 bn net expansion since 2020 corresponds, at the estimated sensitivities, to a persistent downward pressure of several basis points per month of typical issuance on 3-month bills. Second, the redemption asymmetry means that the same sector can withdraw demand far faster than it supplied it: the outflow coefficient is about six times the inflow coefficient in magnitude, so run dynamics are inherently non-linear in their market impact. Third, the connectedness results show that the stablecoin–bond coupling has been strengthening since 2023 and that the sector flips to a net volatility transmitter precisely when the bond market is least able to absorb shocks.

估计结果包含三点经济信息。第一，稳定币对收益率的压缩足迹真实存在，但单位流量的效应温和且集中于储备实际配置的国库券期限；然而就行业整体增长而言，其对短端需求的累计贡献是可观的：2020 年以来的 2,580 亿美元净扩张，按估计的敏感度折算，相当于在典型发行月份对 3 个月期国库券施加每月数个基点的持续下行压力。第二，赎回非对称性意味着该行业撤回需求的速度可以远快于其供给需求的速度：流出系数的量级约为流入系数的六倍，因此挤兑动态的市场影响在本质上是非线性的。第三，连通性结果表明，稳定币与债市的耦合自 2023 年以来持续增强，且该部门恰恰在债市吸收冲击能力最弱的时候翻转为波动净输出者。

For policy, the results cut both ways for the GENIUS Act [10]. On the one hand, mandating high-quality liquid reserve assets reduces the probability that any single issuer's portfolio triggers distress, and the event evidence suggests that reserve-quality commitments are associated with lower bill yields and manageable spread responses. On the other hand, concentrating the entire sector's reserves in short Treasuries deepens the structural coupling documented here: our stress calibrations imply that a system-wide run would transmit into the bill market with three-digit basis-point impacts, a channel that did not exist when reserves were diversified into commercial paper. A prudent calibration would therefore pair reserve-composition mandates with redemption-gate or liquidity-buffer requirements sized to the asymmetric outflow elasticities estimated here, and with monitoring of the SOFR–ON-RRP spread and short-tenor bill dislocations as early-warning indicators of stablecoin stress.

对政策而言，本文结果对 GENIUS Act [10] 具有双向含义。一方面，强制配置高质量流动性储备资产降低了单一发行方组合触发困境的概率，且事件证据表明储备质量承诺与更低的国库券收益率和可控的利差反应相伴。另一方面，将整个行业的储备集中于短期国债加深了本文记录的结构性耦合：压力校准显示系统性挤兑将以三位数基点量级的冲击传导至国库券市场——这一渠道在储备分散于商业票据的时代并不存在。因此，审慎的监管安排应把储备结构强制要求与赎回闸门或流动性缓冲要求相结合，缓冲规模应比照本文估计的非对称流出弹性设定，并将 SOFR–ON-RRP 利差与短期国库券定价错位作为稳定币压力的早期预警指标加以监测。

8. Conclusion / 结论

This paper provides a four-dimensional empirical account of how U.S.-dollar stablecoins affect the bond market. Local projections show that net issuance compresses the 1- and 3-month Treasury yields by up to 3.4 and 2.4 basis points per billion dollars of five-day flow, with no significant effect beyond the two-year maturity. Diebold–Yilmaz and rolling connectedness measures show moderate average spillovers that intensify over time and flip the stablecoin market into a net transmitter in crisis windows. Event studies around the USDC depeg and regulatory milestones document money-market stress and flight-to-quality responses that regressions on daily spreads cannot isolate. Calibrated stress tests imply that a 10% USDT run would raise the 3-month yield by 47–60 basis points and that a \$50 bn joint run would raise it by 126–161 basis points. The overarching message is that stablecoins are now a structural, state-contingent participant in the world's safest market: their growth compresses bill yields in quiet times, and their fragility transmits into bill-market stress in crises. Regulation that makes individual coins safer while coupling the sector ever more tightly to Treasuries should be designed with that trade-off in mind.

本文从四个维度实证刻画了美元稳定币如何影响债券市场。局部投影显示，每 10 亿美元 5 日净流量使 1 个月与 3 个月期国债收益率最多分别下行 3.4 与 2.4 个基点，2 年期以上无显著效应。Diebold–Yilmaz 与滚动连通性测度显示，平均溢出中等但随时间增强，且稳定币市场在危机窗口翻转为净输出者。围绕 USDC 脱锚与监管里程碑的事件研究记录了日度利差回归无法分离的货币市场压力与避险响应。校准压力测试显示，10% 的 USDT 挤兑将推高 3 个月期收益率 47–60 个基点，500 亿美元联合挤兑将推高 126–161 个基点。总体信息是：稳定币已成为全球最安全市场中结构性的、状态依存的参与者——其增长在平静时期压缩国库券收益率，其脆弱性在危机时期向国库券市场传导压力。使个体稳定币更安全、同时将该行业与国债更紧密耦合的监管设计，应当充分权衡这一取舍。

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